Hints: Methods of proof – 4. Invariants, monovariants and double counting

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Look at the parity of the sum of the numbers on the board: replacing a, b by $|a-b|$ changes the sum by $2\min(a,b)$.

Hint 2. So the parity of the sum is invariant, and the last number has the parity of $1+\cdots+n = \frac{n(n+1)}{2}$. Zero is possible only if $n \equiv 0$ or $3 \pmod 4$.

Hint 3. Construction: four consecutive numbers $k, k+1, k+2, k+3$ give $1, 1$ and then 0; for $n \equiv 3$ first turn $1, 2, 3$ into $3-2=1$ and $1-1=0$. Then combine the zeros ($|0-0|=0$).

Problem 2

Hint 1. Colour the cell (i,j), $0 \leqslant i, j \leqslant 9$, with colour $(i+j) \bmod 4$. Every 1×4 piece covers each colour exactly once.

Hint 2. Count the colours on the 10×10 board: they occur $25, 26, 25, 24$ times (colours $0, 1, 2, 3$), not equally often, so no tiling.

Hint 3. General case: an $m \times n$ board is tileable by $1\times k$ pieces iff $k \mid m$ or $k \mid n$. "If" is clear. For "only if", put the weight ω^{i+j} on the cell (i,j), $\omega = e^{2\pi i/k}$: each piece has total weight 0, while the board has weight $\left(\sum_{i<m}\omega^i\right)\left(\sum_{j<n}\omega^j\right)$, which vanishes only if $k \mid m$ or $k \mid n$.

Problem 3

Note. "Changes sides" is read as: the person turns all his friendships into enmities and vice versa (non-friends are enemies). For the reading with two camps and enemies inside one's own camp the same monovariant (hostile pairs inside camps) works.

Hint 1. Watch the total number of hostile pairs.

Hint 2. If the person has e enemies and f friends with $e > f$, after the switch he has f enemies and e friends, and no other pair changes. So the number of hostile pairs drops by $e - f \geqslant 1$.

Hint 3. It is a non-negative integer at most $\binom{n}{2}$ at the start, so there are at most $\binom{n}{2}$ steps.

Problem 4

Hint 1. $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$: choose n people from n men and n women, and sort by the number k of men; use $\binom{n}{k} = \binom{n}{n-k}$.

Hint 2. $\sum_k k\binom{n}{k} = n2^{n-1}$: count committees with a chairman from n people, choosing the committee first or the chairman first. A second way: differentiate $(1+x)^n = \sum \binom{n}{k}x^k$ and put $x = 1$.

Hint 3. Vandermonde $\sum_k \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$: an r-subset of $M \sqcup N$ ($|M| = m$, $|N| = n$) corresponds bijectively to the pair (its part in M, its part in N); sort by the size k of the first part.

Problem 5

Hint 1. Let the design have v points, b blocks of size $k < v$, every point in r blocks and every pair of points in $\lambda \geqslant 1$ blocks. Counting pairs (x, A) with $x \in A$ gives $bk = vr$; counting triples (x, y, A), $x \neq y$ in A, with x fixed gives $r(k-1) = \lambda(v-1)$.

Hint 2. Fisher's inequality $b \geqslant v$ is equivalent to $r \geqslant k$. Fix a block B and let $m_C = |B \cap C|$ for the other $b-1$ blocks. Double count again: $\sum_C m_C = k(r-1)$ and $\sum_C m_C(m_C-1) = k(k-1)(\lambda-1)$.

Hint 3. Cauchy–Schwarz $(\sum m_C)^2 \leqslant (b-1)\sum m_C^2$; after substituting λ and b from Hint 1 it simplifies to $\frac{r(r-k)(v-k)^2}{v-1} \geqslant 0$, so $r \geqslant k$ and $b \geqslant v$. (The usual alternative is the rank of $NN^T = (r-\lambda)I + \lambda J$.)

Problem 6

Hint 1. Trace and characteristic polynomial are invariant under similarity. Over $\mathbb{C}$ (or the algebraic closure) a nilpotent N is similar to a strictly upper triangular matrix, since its only eigenvalue is 0.

Hint 2. Hence $\operatorname{tr} N = 0$ (the sum of the eigenvalues with multiplicity, all zero). For the commutator use $\operatorname{tr}(AB) = \operatorname{tr}(BA)$: $\operatorname{tr}(AB - BA) = 0$, while $\operatorname{tr} E = n \neq 0$ in characteristic 0.

Hint 3. (In characteristic p with $p \mid n$ the trace argument fails, and indeed $AB - BA = E$ is then possible, e.g. for $n = p$ with differentiation and multiplication by x on $\mathbb{F}_p[x]/(x^p)$.)

Problem 7

Hint 1. First check that swapping two adjacent entries changes the number of inversions by exactly ± 1.

Hint 2. A transposition of the entries in positions $i < j$ is a composition of $2(j-i)-1$ adjacent swaps, an odd number; so the parity of the number of inversions flips. Hence $\operatorname{sgn}\colon S_n \to \{\pm 1\}$ is a well-defined homomorphism.

Hint 3. For $n \geqslant 2$ it is surjective, so its kernel A_n has index 2. Application: the 15-puzzle (class work, problem 1), or the Rubik's cube fact that a single swap of two edge pieces is impossible.

Problem 8

Hint 1. Extremal proof: if the set has at least two points, take two points A, B at the minimal distance. What about A and the midpoint of AB?

Hint 2. The midpoint M is in the set and $|AM| = |AB|/2 < |AB|$: contradiction. Monovariant (descent) proof: starting from $A \neq B$, the points $A + (B-A)/2^k$, $k = 0, 1, 2, \ldots$, all belong to the set and are pairwise distinct, contradicting finiteness.

Hint 3. Lattice comparison: $\mathbb{Z}^2$ is not midpoint-closed, but among any five lattice points two have the same parity pattern of coordinates, so their midpoint is a lattice point; there the obstruction is a residue (pigeonhole on $\mathbb{Z}^2/2\mathbb{Z}^2$) instead of a size.

Problem 9

Hint 1. If n has at least two digits, then $S(n) < n$: compare $\sum d_i 10^i$ with $\sum d_i$. So the value strictly decreases until it is a single digit.

Hint 2. Since $10^i \equiv 1 \pmod 9$, $n \equiv S(n) \pmod 9$: the residue modulo 9 is invariant.

Hint 3. The process ends at the unique digit $1, \ldots, 9$ congruent to n modulo 9: the digital root of $n \geqslant 1$ is $1 + ((n-1) \bmod 9)$.

Problem 10

Hint 1. Take a connected plane graph. If it contains a cycle, delete one edge of the cycle. What happens to V, E and F?

Hint 2. The cycle separates the plane (Jordan curve theorem), so the edge borders two different faces, which merge: E and F both drop by 1 and $V - E + F$ is unchanged. E is a decreasing monovariant, so the process ends in a tree, where $V - E + F = V - (V - 1) + 1 = 2$.

Hint 3. K_5: if it were planar, $F = 2 - 5 + 10 = 7$. Count edge–face incidences: each face is bounded by at least 3 edges, each edge borders at most 2 faces, so $3F \leqslant 2E = 20$, i.e. $F \leqslant 6$. Contradiction.